

系统 I 习题

1-1

Convert the following numbers from the given base to the other three bases listed in the table:

Decimal	Binary	Octal	Hexadecimal
369.3125	101110001.0101	561.24	171.5
189.625	10111101.101	275.5	BD.A
214.625	11010110.101	326.5	D6.A
62407.625	1111001111000111.101	171707.5	F3C7.A

1-2

In each of the following cases, determine the radix r :

$$(a) (BEE)_r = (2699)_{10}$$

$$(b) (365)_r = (194)_{10}$$

(a)

$$11r^2 + 14r + 14 = 2699, \text{ so } r = 15.$$

(b)

$$3r^2 + 6r + 5 = 194, \text{ so } r = 7.$$

2-1

For the Boolean functions E and F , as given in the following truth table:

X	Y	Z	E	F
0	0	0	0	1
0	0	1	1	0
0	1	0	1	1
0	1	1	0	0
1	0	0	1	1
1	0	1	0	0
1	1	0	1	0
1	1	1	0	1

- List the minterms and maxterms of each function.
- List the minterms of $\overline{E}$ and $\overline{F}$
- List the minterms of $E + F$ and $E \cdot F$.
- Express E and F in sum-of-minterms algebraic form.
- Simplify E and F to expressions with a minimum of literals.

(a)

Minterms:

$$E = m(1, 2, 4, 6)$$

$$F = m(0, 2, 4, 7)$$

Maxterms:

$$E = M(0, 3, 5, 7)$$

$$F = M(1, 3, 5, 6)$$

(b)

$$\overline{E} = m(0, 3, 5, 7)$$

$$\overline{F} = m(1, 3, 5, 6)$$

(c)

$$E + F = m(0, 1, 2, 4, 6, 7)$$

$$E \cdot F = m(2, 4)$$

(d)

$$E = \overline{X}\overline{Y}Z + \overline{X}Y\overline{Z} + X\overline{Y}\overline{Z} + XY\overline{Z}$$

$$F = \overline{X}Y\overline{Z} + \overline{X}Y\overline{Z} + X\overline{Y}\overline{Z} + XYZ$$

(e)

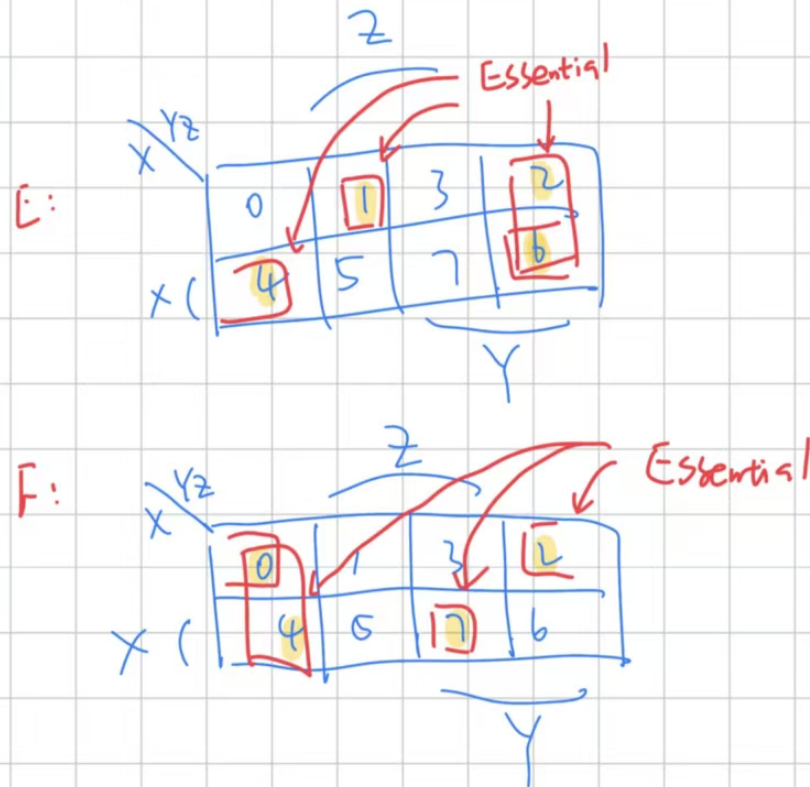
$$E = \overline{X}Y\overline{Z} + Y\overline{Z} + X\overline{Z} = \overline{X}Y\overline{Z} + \overline{Z}(X + Y)$$

$$F = \overline{Y}\overline{Z} + \overline{X}\overline{Z} + XYZ = \overline{Z}(\overline{Y} + \overline{X}) + XYZ$$



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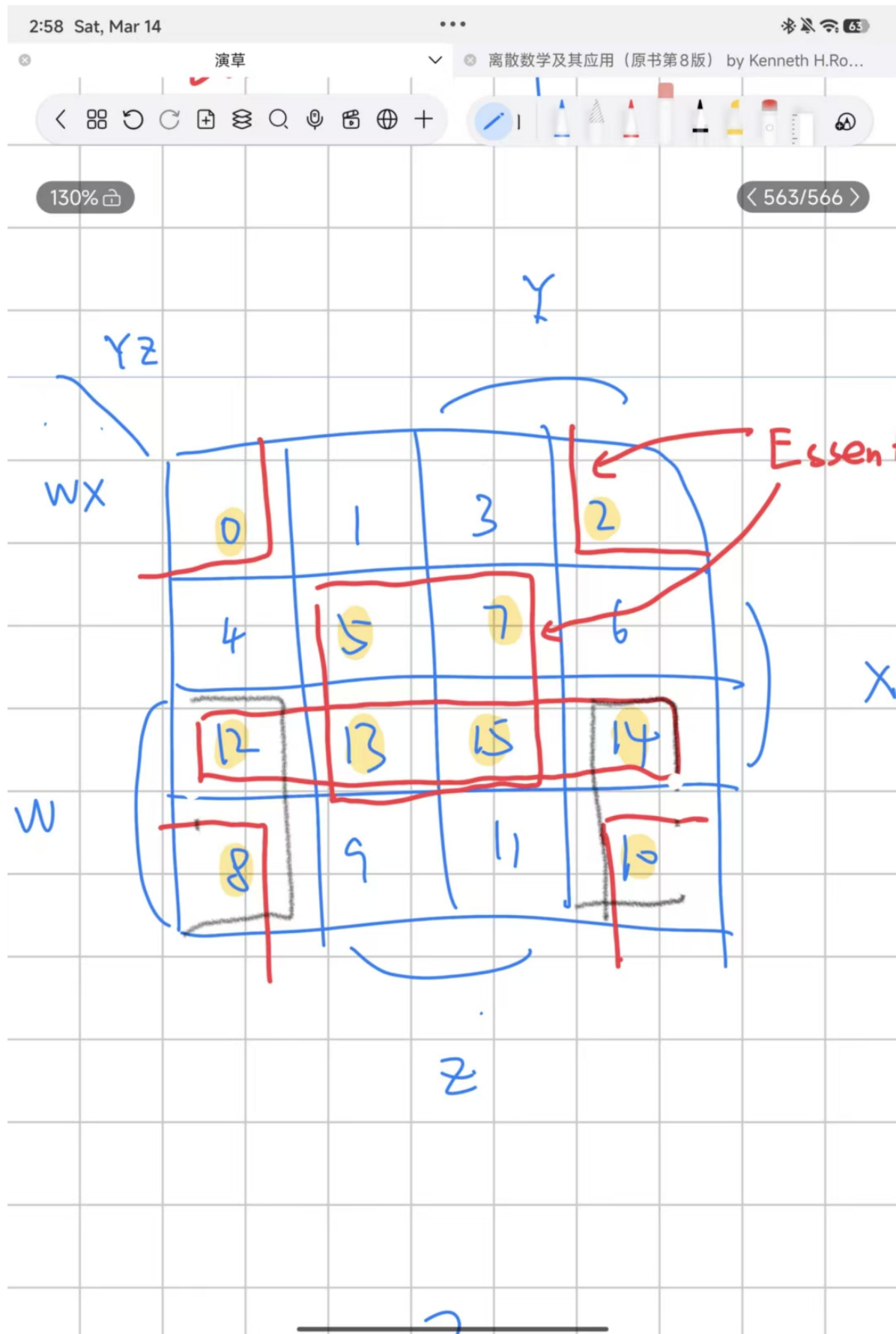
2-2

Find all the prime implicants for the following Boolean functions, and determine which are essential:

$$F(W, X, Y, Z) = \sum m(0, 2, 5, 7, 8, 10, 12, 13, 14, 15)$$

prime implicants: $m(0, 2, 8, 10), m(5, 7, 13, 15), m(12, 13, 15, 14), m(12, 8, 14, 10)$

essential: $m(0, 2, 8, 10), m(5, 7, 13, 15)$



2-3

Prove the identity of the following Boolean equation (a), using algebraic manipulation (只做 a):

(a)

$$ABC\overline{C} + B\overline{C}\overline{D} + BC + \overline{C}D = B + \overline{C}D$$

(b)

$$WY + \overline{W}YZ + WXZ + \overline{W}XY\overline{Z} = WY + \overline{W}XZ + \overline{X}YZ + XY\overline{Z}$$

(c)

$$A\overline{D} + \overline{A}B + \overline{C}D + \overline{B}C = (\overline{A} + \overline{B} + \overline{C} + \overline{D})(A + B + C + D)$$

(a)

$$\begin{aligned} ABC\overline{C} + B\overline{C}\overline{D} + BC + \overline{C}D &= ABC\overline{C}(D + \overline{D}) + B\overline{C}\overline{D} + BC + \overline{C}D \\ &= ABC\overline{C}D + B\overline{C}\overline{D} + BC + \overline{C}D \\ &= B\overline{C}\overline{D} + BC + \overline{C}D \\ &= B\overline{C}\overline{D} + BC(D + \overline{D}) + \overline{C}D(B + 1) \\ &= B\overline{C}\overline{D} + BCD + BC\overline{D} + B\overline{C}D + \overline{C}D \\ &= B + \overline{C}D \end{aligned}$$

2-4

Optimize the following Boolean functions F together with the don't-care conditions d . Find all prime implicants and essential prime implicants, and apply the selection rule.

$$F(W, X, Y, Z) = \Sigma m(0, 2, 4, 5, 8, 14, 15), \quad d(W, X, Y, Z) = \Sigma m(7, 10, 13)$$

prime implicants: $m(0, 2, 8, 10), m(5, 7, 13, 15), m(0, 4), m(4, 5), m(15, 14), m(14, 10)$

essential prime implicants: $m(0, 2, 8, 10)$

selective ones: $m(4, 5), m(15, 14)$

2:13 Sat, Mar 14

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Hand-drawn Karnaugh map for a 4-variable function with variables W, X, Y, and Z. The map is a 4x4 grid with cells numbered 0 to 15. Red lines group the cells into prime implicants. The groups are: (0, 1, 2, 3) labeled 'Essential', (4, 5, 6, 7), (8, 9, 10, 11), (12, 13, 14, 15), (0, 4, 8, 12), (1, 5, 9, 13), (2, 6, 10, 14), (3, 7, 11, 15), (0, 1, 4, 5), (2, 3, 6, 7), (8, 9, 12, 13), (10, 11, 14, 15), (0, 4, 8, 12), (1, 5, 9, 13), (2, 6, 10, 14), (3, 7, 11, 15).

3-1

A hierarchical component with the function

$$H = \bar{X}Y + XZ$$

is to be used along with inverters to implement the following equation:

$$G = \bar{A}\bar{B}C + \bar{A}BD + A\bar{B}\bar{C} + AB\bar{D}$$

The overall circuit can be obtained by using Shannon's expansion theorem,

$$F = \bar{X}F_0(X) + XF_1(X)$$

where $F_0(X)$ is F evaluated with variable $X = 0$ and $F_1(X)$ is F evaluated with variable $X = 1$. This expansion F can be implemented with function H by letting $Y = F_0$ and $Z = F_1$. The expansion theorem can then be applied to each of F_0 and F_1 using a variable in each, preferably one that appears in both true and complemented form. The process can then be repeated until all F_i 's are single literals or constants.

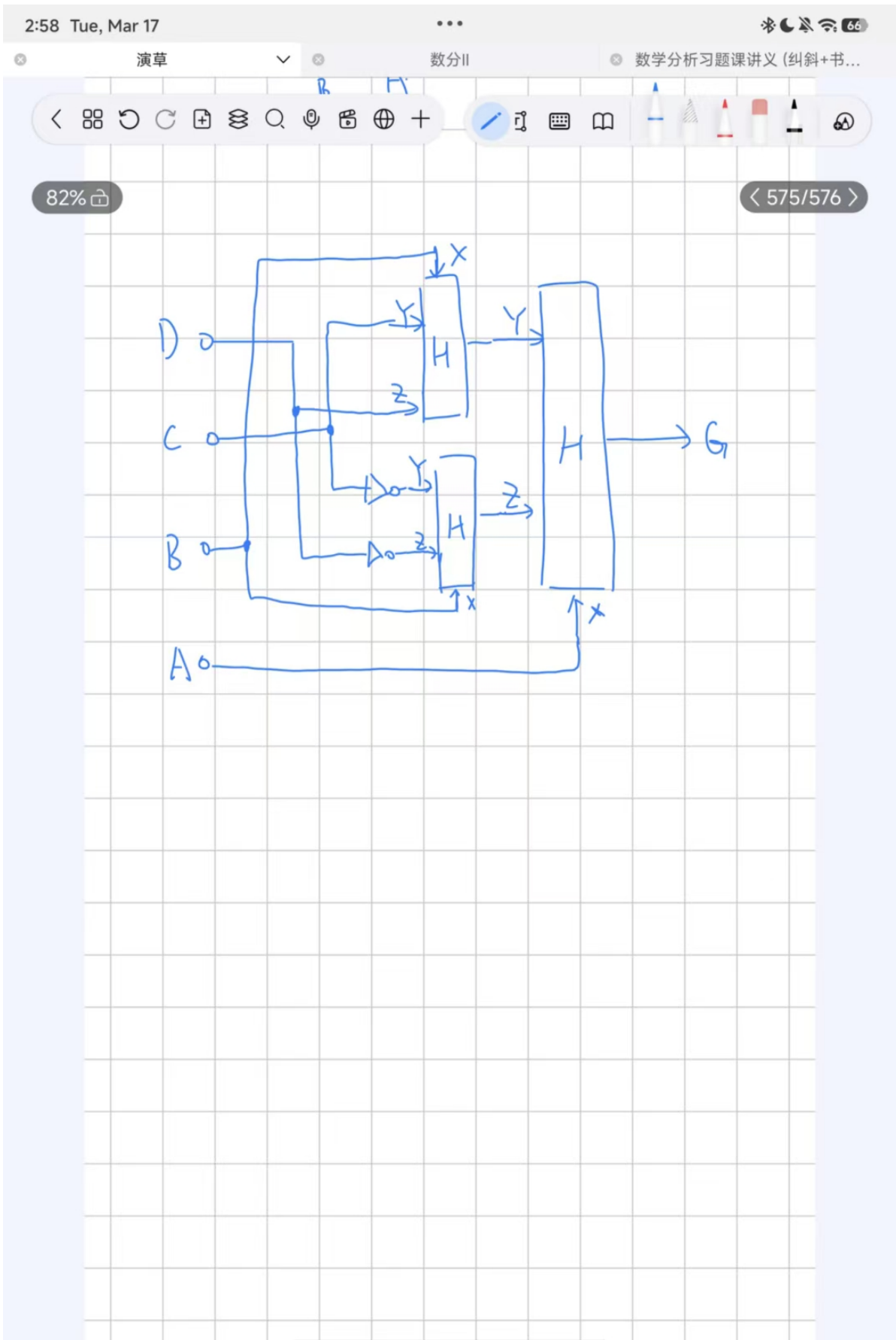
For G , use $X = A$ to find G_0 and G_1 and then use $X = B$ for G_0 and G_1 . Draw the top-level diagram for G using H as a hierarchical component.

$$G = \bar{A}(\bar{B}C + BD) + A(\bar{B}\bar{C} + B\bar{D})$$

$$\text{So } G_0 = \bar{B}C + BD, G_1 = \bar{B}\bar{C} + B\bar{D}$$

Applying Shannon's expansion to G_0, G_1 won't change the way they look.

Top-level diagram:

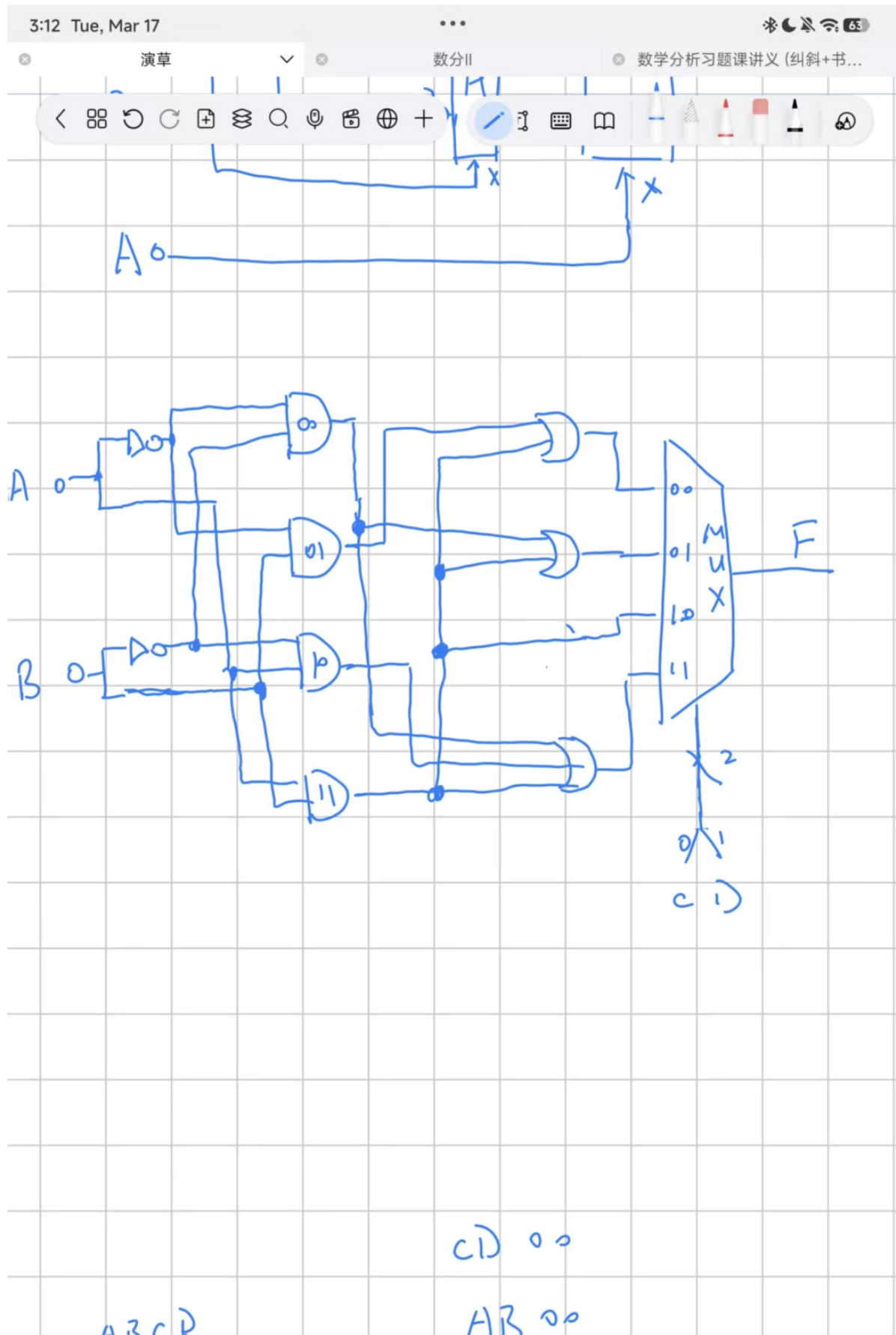


3-2

Implement the Boolean function

$$F(A, B, C, D) = \sum m(1, 3, 4, 11, 12, 13, 14, 15)$$

with a 4-to-1-line multiplexer and external gates. Connect inputs A and B to the selection lines. The input requirements for the four data lines will be a function of the variables C and D. The values of these variables are obtained by expressing F as a function of C and D for each of the four cases when $AB = 00, 01, 10, \text{ and } 11$. These functions must be implemented with external gates.



3:12 Tue, Mar 17

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数分II

数学分析习题课讲义 (纠错+书...

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ABC	P
0 000	0
1 001	0
2 010	0
3 011	0
4 100	0
5 101	0
6 110	0
7 111	0
8 100	0
9 101	0
10 110	0
11 111	0
12 110	0
13 110	0
14 110	0
15 111	0

AB	CD
00	00
01	01
10	10
11	11
00	01
01	00
10	10
11	11
00	10
01	01
10	10
11	11
00	01
01	00
10	10
11	11

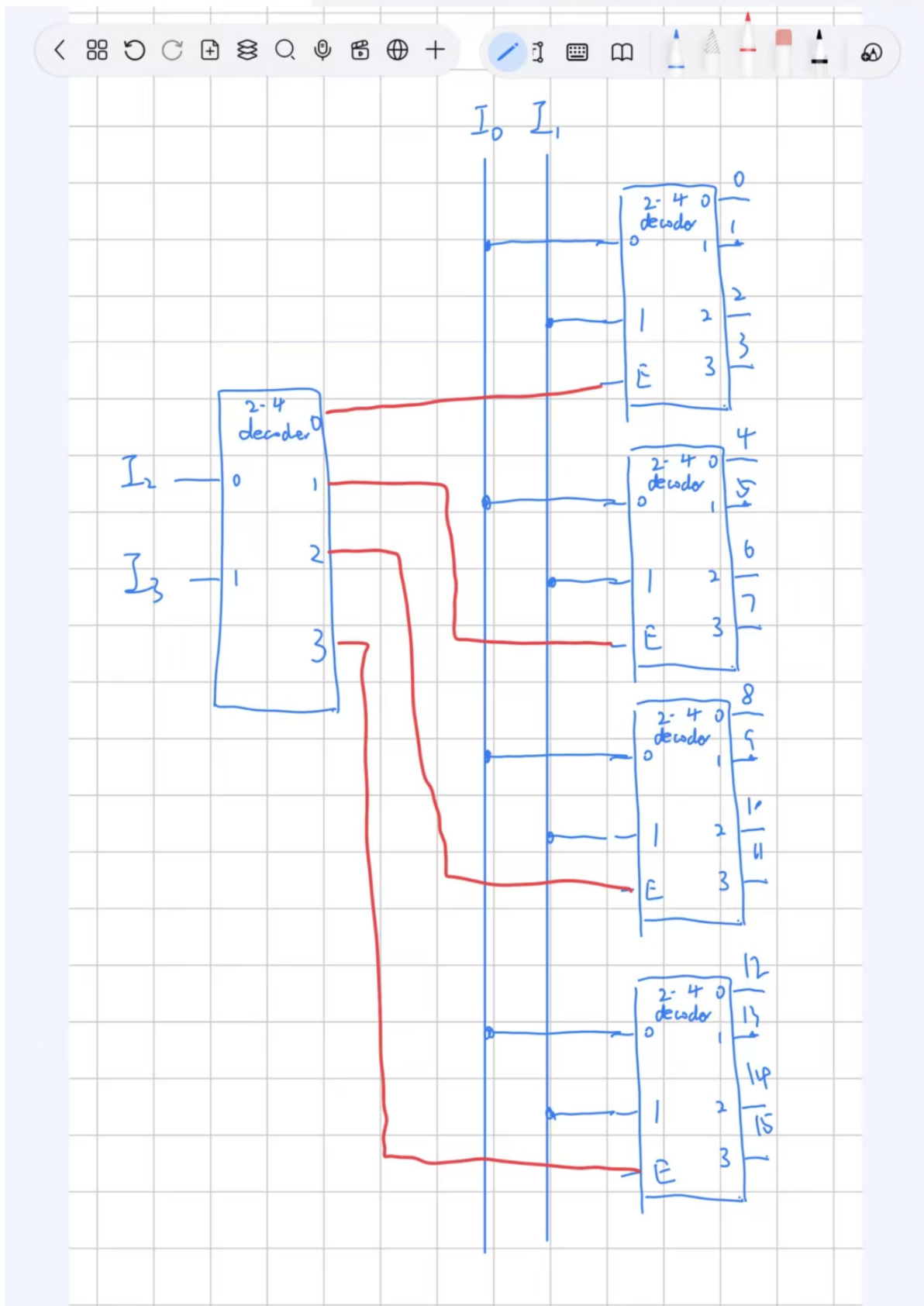
3-3

- (a) Design a 4-to-16-line decoder using two 3-to-8-line decoders and 16 2-input AND gates.
 (b) Design a 4-to-16-line decoder with enable using five 2-to-4-line decoders with enable.

(a)

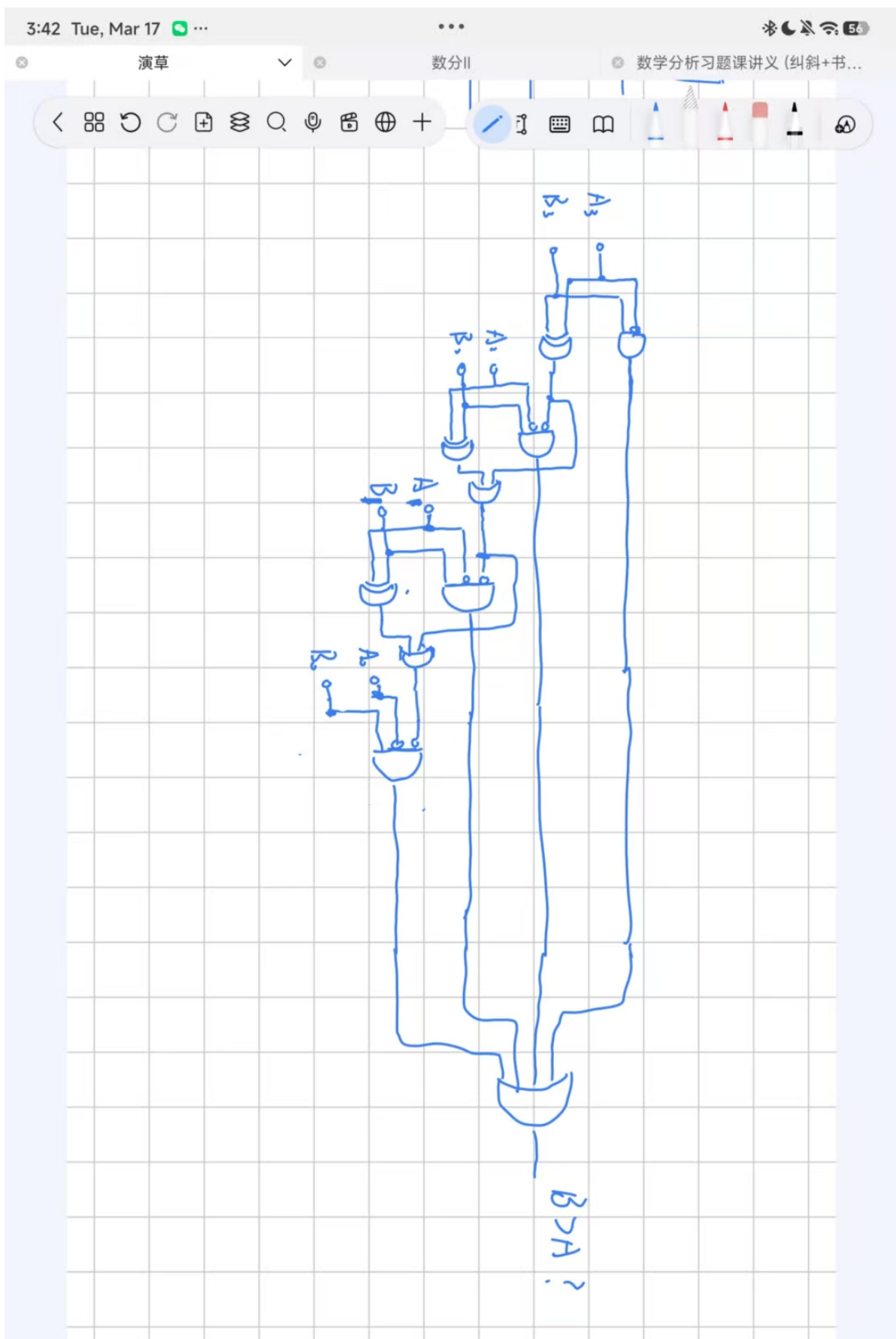


(b)



3-4

Design a combinational circuit that compares two 4-bit unsigned numbers A and B to see whether B is greater than A . The circuit has one output X , so that $X = 1$ if $A < B$ and $X = 0$ if $A \geq B$.



4-1

3.20

< 3.5 > What decimal number does the bit pattern $0x0C'000000$ represent if it is a two's complement integer? An unsigned integer?

Under both conditions, the number is 201326592.

< 3.5 > If the bit pattern 0x0C000000 is placed into the Instruction Register, what MIPS instruction will be executed?

So it's `jal 0x00000000`

So it's a normalized number, $(-1)^0 \times (1 + 0.0) \times 2^{24-127} = 2^{-103}$

So the binary representation is 01000010011110100000000000000000.

01000000010011110100.

4-2

In comparison, the single precision range $\approx 3.4 \times 10^{38}$, precision contains $23 \log_{10} 2 \approx 7$ decimal digits.

5-1

A sequential circuit has one flip-flop Q, two inputs X and Y, and one output S. The circuit consists of a D flip-flop with S as its output and logic implementing the function $D = X \oplus Y \oplus S$ with D as the input to the D flip-flop. Derive the state table and state diagram of the sequential circuit.

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Next state $A(t+1) = D(t)$

Output $S(t) = A(t)$

Input $D(t) = X(t) \oplus Y(t) \oplus S(t)$

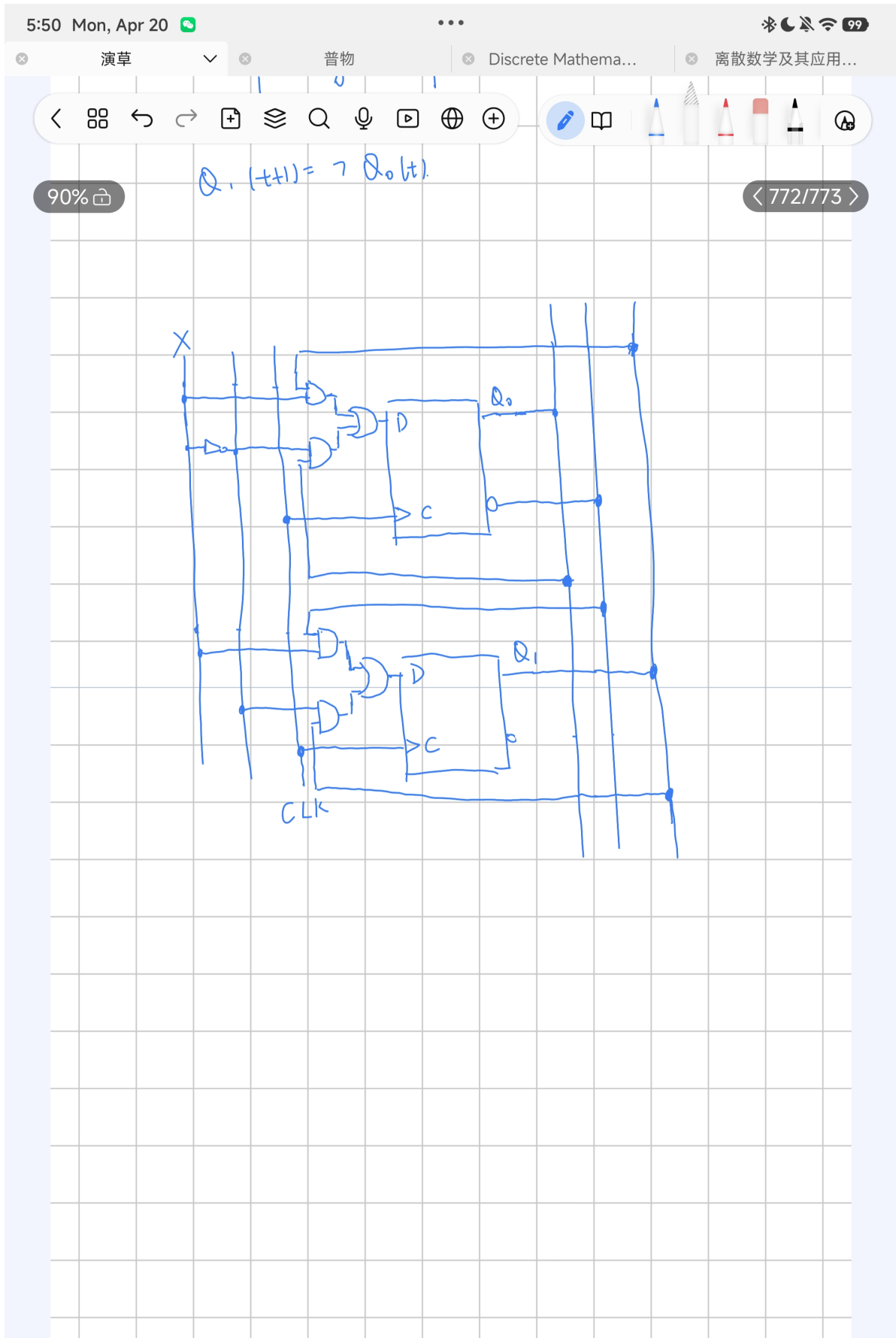
Present State $A(t)$	Next state				Output			
	$X(t)=0, Y(t)=0$ $A(t+1)$	$X(t)=0, Y(t)=1$ $A(t+1)$	$X(t)=1, Y(t)=0$ $A(t+1)$	$X(t)=1, Y(t)=1$ $A(t+1)$	$X(t)=0, Y(t)=0$ $S(t)$	$X(t)=0, Y(t)=1$ $S(t)$	$X(t)=1, Y(t)=0$ $S(t)$	$X(t)=1, Y(t)=1$ $S(t)$
0	0	1	0	1	0	0	0	0
1	1	0	1	0	1	1	1	1

```

graph LR
    00((0/0)) -- "X=0, Y=0" --> 00
    00 -- "X=0, Y=1" --> 11((1/1))
    00 -- "X=1, Y=0" --> 11
    00 -- "X=1, Y=1" --> 00
    11 -- "X=0, Y=0" --> 00
    11 -- "X=0, Y=1" --> 00
    11 -- "X=1, Y=0" --> 00
    11 -- "X=1, Y=1" --> 11
  
```

5-2

Design a sequential circuit with two D flip-flops A and B and one input X. When $X = 0$, the state of the circuit remains the same. When $X = 1$, the circuit goes through the state transitions from 00 to 10 to 11 to 01, back to 00, and then repeats.



A pair of signals Request (R) and Acknowledge (A) is used to coordinate transactions between a CPU and its I/O system. The interaction of these signals is often referred to as a "handshake." These signals are synchronous with the clock and, for a transaction, are to have their transitions always appear in the order shown in Figure 4-53. A handshake checker is to be designed that will verify the transition order. The checker has inputs, R and A , asynchronous reset signal, RESET, and output, Error (E). If the transitions in a handshake are in order, $E = 0$. If the transitions are out of order, then E becomes 1 and remains at 1 until the asynchronous reset signal (RESET = 1) is applied to the CPU.

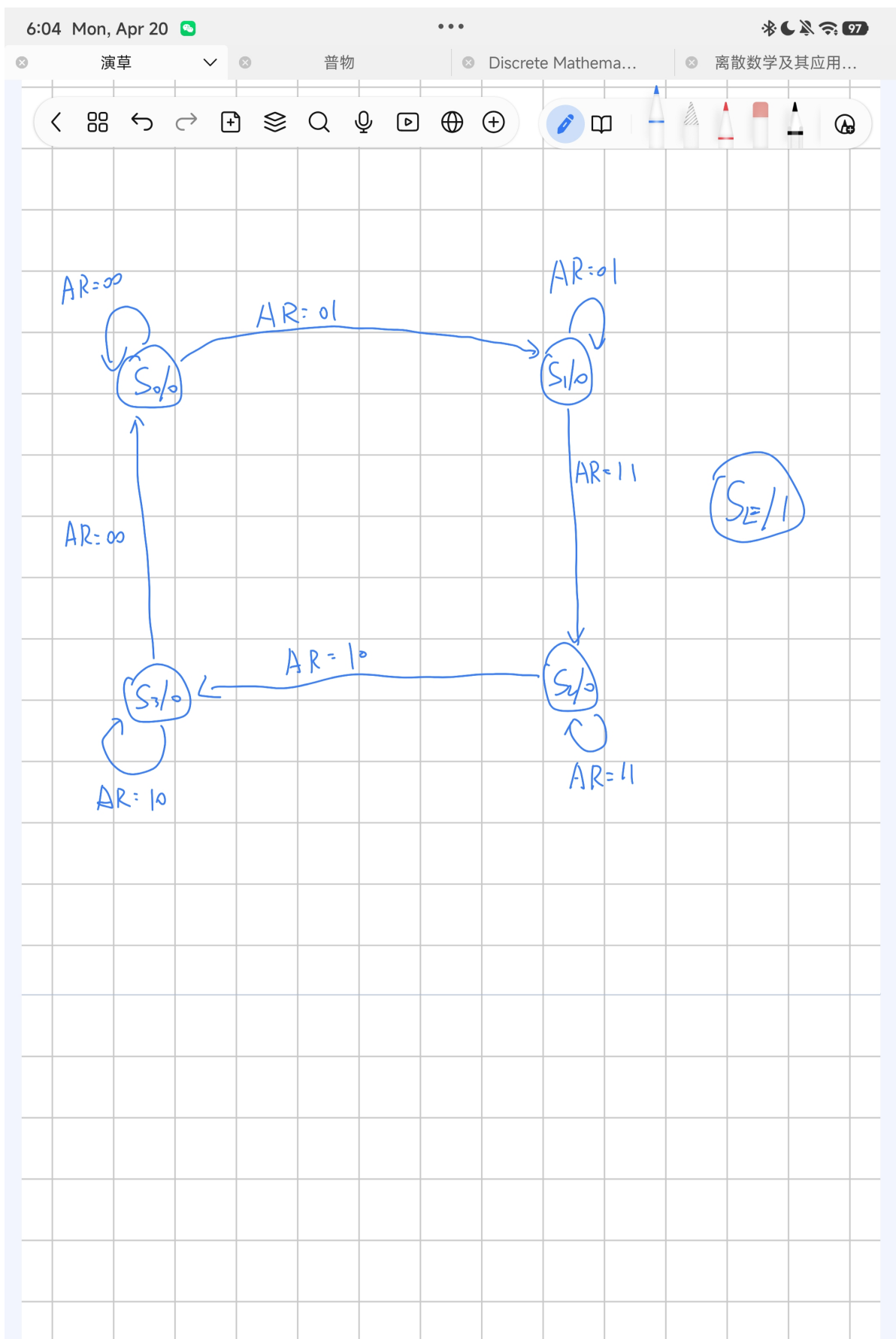
(a) Find the state diagram for the handshake checker.

(b) Find the state table for the handshake checker.

(a)

If RESET=1, the next state will be S0 (not drawn in diagram).

All other edges not drawn are connected to SE.



(b)

6:09 Mon, Apr 20

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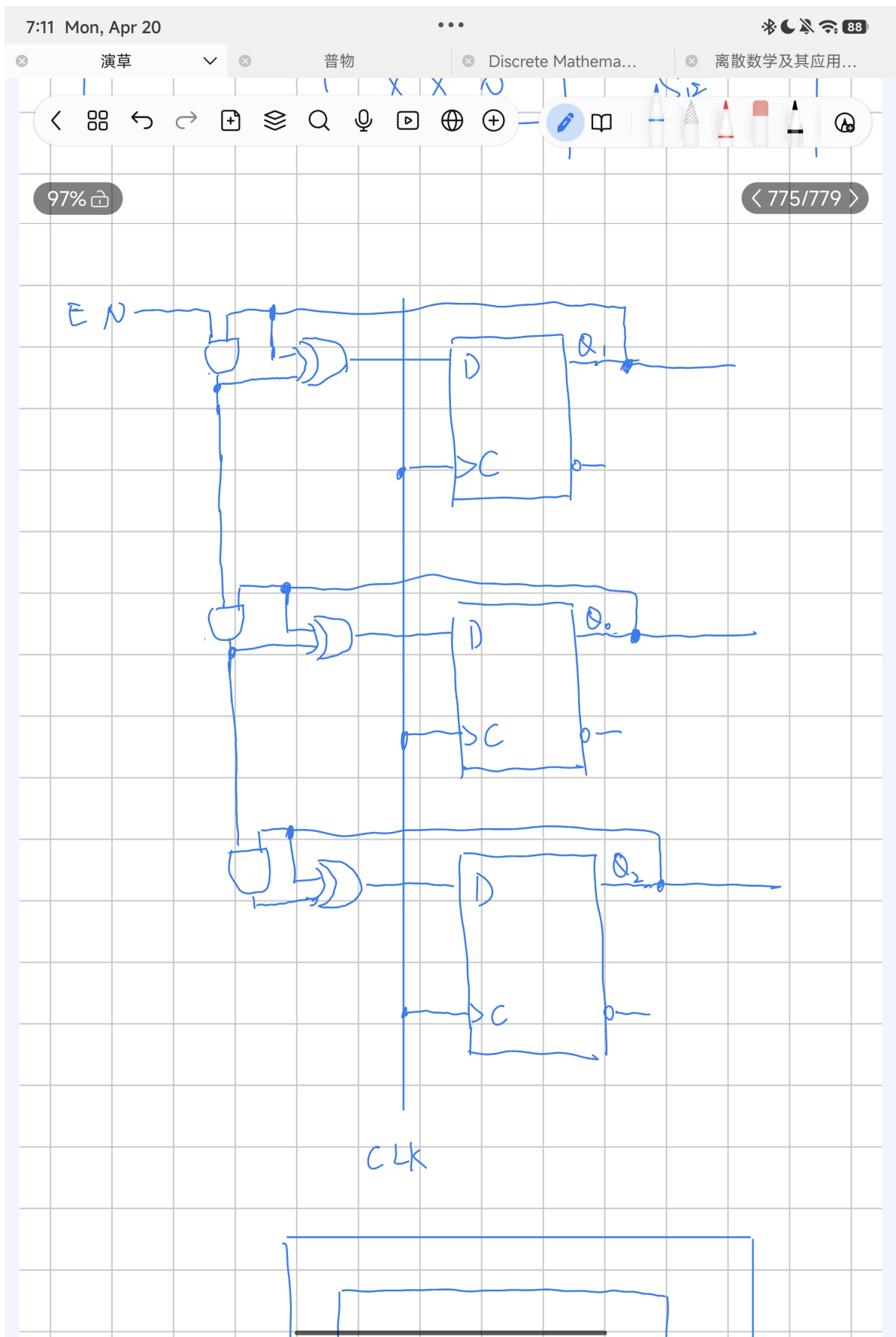
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Present state	Input				Next state
	A	R	RST		
S ₀	X	X	1		S ₀
	0	0	0		S ₀
	0	1	0		S ₁
	X	X	0		S _E
S ₁	X	X	1		S ₀
	0	1	0		S ₁
	1	1	0		S ₂
	X	X	0		S _E
S ₂	X	X	1		S ₀
	1	1	0		S ₂
	1	0	0		S ₃
	X	X	0		S _E
S ₃	X	X	1		S ₀
	1	0	0		S ₃
	0	0	0		S ₀
	X	X	0		S _E

5-4

Use *D*-type flip-flops and gates to design a counter with the following repeated binary sequence:
0, 2, 1, 3, 4, 6, 5, 7.

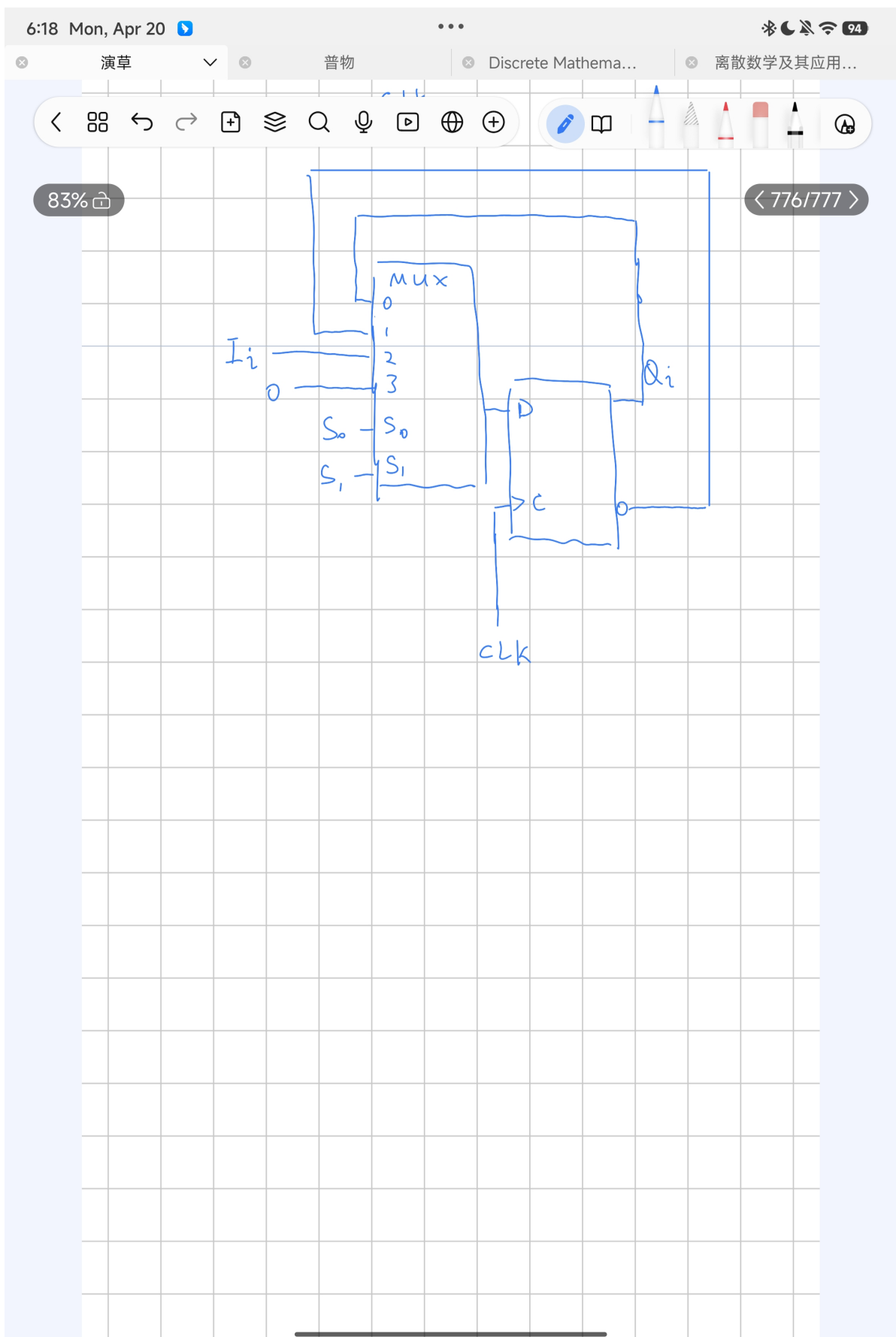


5-5

Draw the logic diagram of a 4-bit register with mode selection inputs S_1 and S_0 . The register is to be operated according to the function table below.

$S1$	$S0$	Register Operation
0	0	No change
0	1	Complement output
1	0	Load parallel data
1	1	Clear register to 0

The structer of other 3 registers is the same.



5-6

A register cell is to be designed for an 8-bit register A that has the following register transfer functions:

$$C_0 : A \leftarrow A \cap B, \quad C_1 : A \leftarrow A \cup \overline{B}$$

Find optimum logic using AND, OR, and NOT gates for the D input to the D flip-flop in the cell.

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$D_i = A_i \cdot \overline{C_0} \cdot \overline{C_1} + A_i B_i C_0 + (A_i + \overline{B_i}) \cdot C_1$

Truth Table:

$A_i B_i \backslash C_0 C_1$	00	01	11	10
00	0	1	X	0
01	0	0	X	0
11	1	1	X	1
10	1	1	X	0

Optimized Logic Expression:

$$D_i = \overline{C_0} A_i + A_i B_i + \overline{B_i} \overline{C_0} C_1$$

Circuit Diagram:

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Handwritten notes and diagrams on a grid background:

Top diagram: A Karnaugh map showing a function with inputs A_i and B_i . The map is divided into four quadrants. The top-left quadrant is labeled C_0 . The top-right quadrant is labeled C_1 . The bottom-left quadrant is labeled C_2 . The bottom-right quadrant is labeled C_3 . The map shows a function that is 1 in the top-left and bottom-right quadrants, and 0 in the top-right and bottom-left quadrants.

Equation:

$$D_i = \overline{C_0} A_i + A_i B_i + \overline{B_i} \overline{C_0} C_1$$

Logic diagram:

The logic diagram implements the equation $D_i = \overline{C_0} A_i + A_i B_i + \overline{B_i} \overline{C_0} C_1$. It features three inputs: C_0 , B_i , and A_i . The output is D_i . The diagram uses three 3-input AND gates and one 3-input OR gate. The inputs to the AND gates are $\overline{C_0}$, A_i , and C_1 ; A_i , B_i , and C_1 ; and $\overline{B_i}$, $\overline{C_0}$, and C_1 . The outputs of the AND gates are connected to the inputs of the OR gate, which produces the final output D_i .

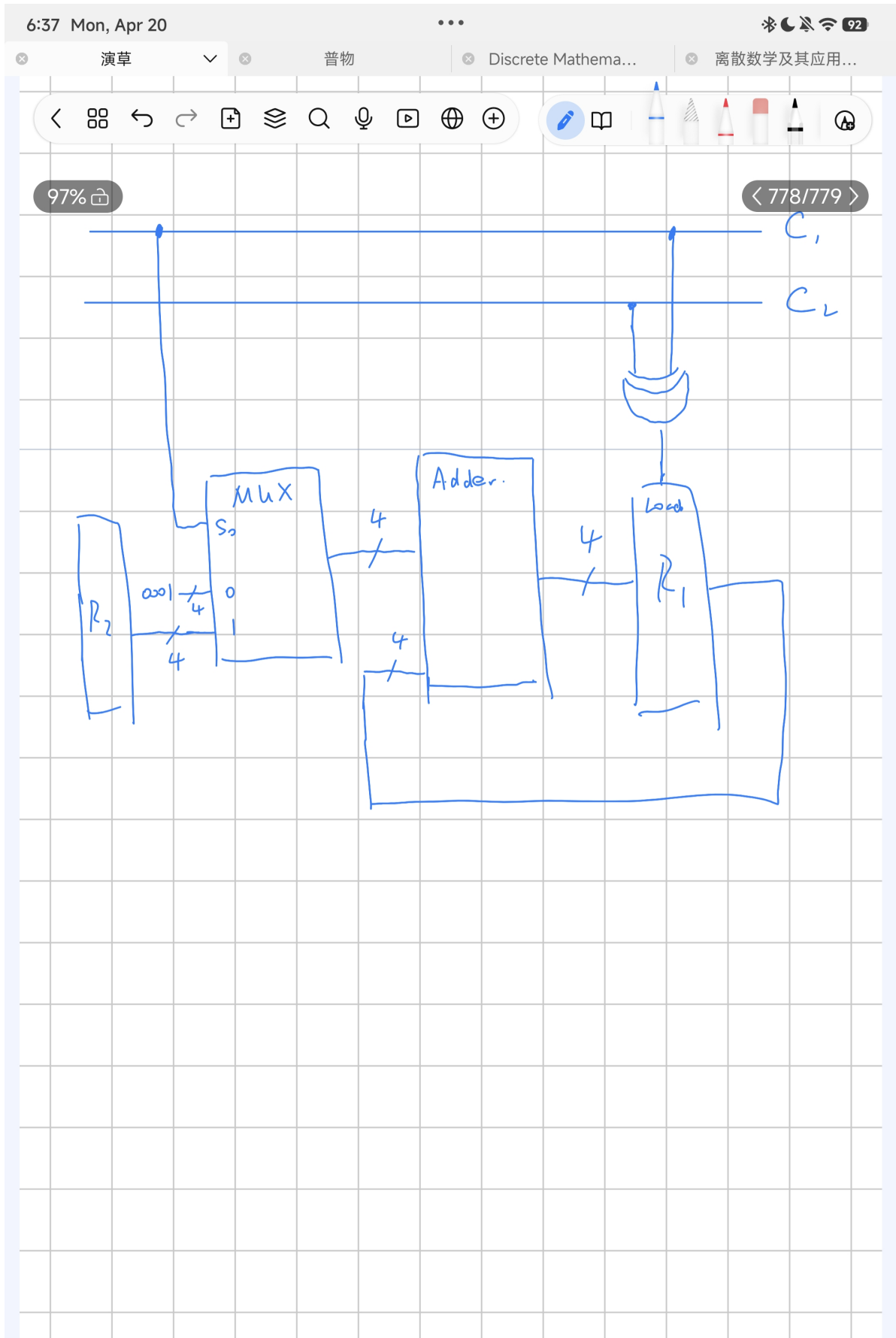
5-7

Two register transfer statements are given (otherwise, R1 is unchanged):

equation[[333, 436, 661, 474]]

(a) Using a 4-bit adder plus external gates as needed, draw the logic diagram that implements these register transfers.

(a)



5-8

A sequential circuit is shown in the following. The timing parameters for the gates and flip-flops are as follows:

- Inverter: $t_{pd} = 0.01$ ns

- XOR gate: $t_{pd} = 0.04 \text{ ns}$
- Flip-flop: $t_{pd} = 0.08 \text{ ns}$, $t_s = 0.02 \text{ ns}$, and $t_h = 0.01 \text{ ns}$

- (a) Find the longest path delay from an external circuit input passing through gates only to an external circuit output.
- (b) Find the longest path delay in the circuit from an external input to positive clock edge.
- (c) Find the longest path delay from positive clock edge to output.
- (d) Find the longest path delay from positive clock edge to positive clock edge.
- (e) Determine the maximum frequency of operation of the circuit in megahertz (MHz).

(a)

X -> XOR -> XOR -> Y: 0.08ns

(b)

X -> XOR -> NOT -> D: 0.05ns

(c)

FF -> A -> XOR -> XOR: 0.16ns

(d)

FF -> A -> XOR -> NOT: 0.13ns

(e)

$$t_{min} \geq t_s + t_{FF \rightarrow FF} = 0.15 \text{ ns}$$

$$f_{max} = \frac{1}{t_{min}} \approx 6666.67 \text{ MHz}$$

6-1

2.12 [5] < § § 2.2, 2.5 > Provide the instruction type and assembly language instruction for the following binary value:

0000 0000 0001 0000 1000 0000 1011 0011_{two}

opcode: 0110011, so R-type.

funct3: 000, funct7: 0000000, so add

rs1: 00001 rs2: 00001 rd: 00001

So it's add x1, x1, x1

6-2

2.13 [5] < § § 2.2, 2.5 > Provide the instruction type and hexadecimal representation of the following instruction:

sd $\times 5, 32(\times 30)$

store, so opcode: 0100011

funct3: 011

rs1: x30, 11110

rs2: x5, 00101

imm: 32, 000000100000

So it's 0000 0010 0101 1111 0011 0000 0010 0011

So 0x025f3023

6-3

2.31 [20] < § 2.8 > Translate function f into RISC-V assembly language. Assume the function declaration for g is `int g(int a, int b)`. The code for function f is as follows:

```
1  int f(int a, int b, int c, int d) {  
2      return g(g(a, b), c + d);  
3  }
```

```
addi sp, -16  
sw ra, 8(sp)  
sw a2, 4(sp)  
sw a3, 0(sp)  
call g  
lw a2, 8(sp)  
lw a3, 0(sp)  
add a1, a2, a3  
call g  
ld ra, 8(sp)  
addi sp, sp, 16  
ret
```

6-4

2.35 Consider the following code:

```
1  lb x6, 0(x7)  
2  sd x6, 8(x7)
```

Assume that the register x7 contains the address 0×10000000 and the data at address is $0 \times 1122334455667788$.

2.35.1 [5] < § 2.3, 2.9 > What value is stored in 0×10000008 on a big-endian machine?

0x11

2.35.2 [5] < § 2.3, 2.9 > What value is stored in 0×10000008 on a little-endian machine?

0x88

6-6

2.36 [5] < § 2.10 > Write the RISC-V assembly code that creates the 64-bit constant $0 \times 1122334455667788_{\text{two}}$ and stores that value to register x10.

```
lui x10, 0x11223  
addi x10, x10, 0x344  
slli x10, 12  
addi x10, x10, 0x556  
slli x10, 12
```

```
addi x10, x10, 0x677
slli x10, x10, 8
addi x10, x10, 0x88
```

7-1

4.13

Examine the difficulty of adding a proposed ss rs1, rs2, imm (Store Sum) instruction to RISC-V.

Interpretation: $\text{Mem}[\text{Reg}[\text{rs1}]] = \text{Reg}[\text{rs2}] + \text{immediate}$

4.13.1 [10] < § 4.4 > Which new functional blocks (if any) do we need for this instruction?

None.

4.13.2 [10] < § 4.4 > Which existing functional blocks (if any) require modification?

Yes. Immgen will need to be modified to satisfy this instruction.

4.13.3 [5] < § 4.4 > What new data paths do we need (if any) to support this instruction?

When this instruction is detected, the following paths have to be added:

From RegFile rs1 to dataMem writeAddr

From RegFile rs2 to ALU input_a

From ALU alu_res to dataMem writeData

4.13.4 [5] < § 4.4 > What new signals do we need (if any) from the control unit to support this instruction?

Yes.

We'll need alu_sel_a to choose from rs1 and rs2

We'll need dmem_writedata_sel to choose from ALU and rs2

We'll need dmem_writeaddr_sel to choose from ALU and rs1

4.13.5 [5] < § 4.4 > Modify Figure 4.21 to demonstrate an implementation of this new instruction.

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Building the Datapath & Controller

□ There are 7+4 signals

